

# Case 1: Moldovan Coffins

McKinsey, 1<sup>st</sup> Round

## About This Case

This case was given by McKinsey in first-round interviews and is a “command and control” case. In this style of case, the interviewer allows the candidate to drive the case initially to explore possible routes to a solution. However, once the candidate has laid out a plan, the interviewer takes control and asks the candidate to solve a few specific problems before coming to the final conclusion.

When giving this case, allow for some initial planning and brainstorming by the candidate, but then firmly take control of each of the “modules” described below. Try to move the candidate along through each of them, since in the actual interview only those candidates that complete all of the sections will be considered to have done well. This case tests mental horsepower and the ability to move to conclusions quickly.

## The Problem

Our client is a coffin maker in the Eastern European country of Moldova. He has seen substantial change in his market in recent years and is contemplating the future of his business. Up until now, he has been in the business of building high-quality, hand-crafted coffins largely by hand with a skilled labor force. Recently, however, he has become aware of technology that would allow him to build machine-made coffins with much less labor. Should he invest in this new technology, and should he even remain in the coffin business in the first place?

## Information Gathering

*Read this information well before you give the case. Note that this case requires you (the interviewer) to drive key points in the discussion. Allow the candidate to formulate a plan and then prompt him/her to consider each of the topics listed below.*

**Prompt: What strategic alternatives should the owner consider?**

*Good Answer: If the candidate doesn't get all of this, help them along since we need to lay this foundation for the rest of the case - We need to decide firstly whether to stay in business at all and if so, whether he uses the new technology:*

- Option 1: Sell the business to a third party
- Option 2: Sell the assets of the company and shut it down
- Option 3: Keep operating as is
- Option 4: Keep operating and invest in the new technology

**Prompt: How would you figure out the current value of the business?**

*Provide the following information if the candidate asks for it clearly and directly.*

**Market Size** – If the candidate asks for the size of the market, first make him/her brainstorm about different ways to determine market size. A good candidate should come up with at least 4 different ways, such as:

- Calculate from population growth, total population, and birth rate
- Review of death records for a period of time
- Take sample of number of obituaries in paper serving given population base
- Calculate from population, average life expectancy

*Now make them calculate the market size, giving them the following data:*

|                               |                                                                          |
|-------------------------------|--------------------------------------------------------------------------|
| <b>Population of Moldova:</b> | 4M                                                                       |
| <b>Population Growth:</b>     | 0%                                                                       |
| <b>Avg Life Expectancy:</b>   | 75 years                                                                 |
| <b>Age Distribution:</b>      | assume a flat age distribution (i.e. same number of people at every age) |
| <b>Burial Customs:</b>        | 75% of deaths are buried in coffins.                                     |

*Right Answer: 40,000 coffins purchased / year. Note that the candidate needs to quickly realize that every year, 1/75<sup>th</sup> of the population will turn 76 and therefore (on average) will die.*

**Price** – Coffins are priced at \$5,000 for a hand-made coffin.

**Costs** – Material accounts for 10% of the direct cost, while labor accounts for the other 90%. COGS is \$4,800 per coffin. Fixed costs for the business are \$700,000 per year. Assume all assets are fully depreciated and ignore taxes.

**Competition** – Moldovan Coffins has a 10% market share and a relative market share of about 1 (*if asked, you may explain that relative market share is the ratio of the company's market share to that of its nearest competitor.*)

**Market Trends, Regulation, etc.** – If asked about any exogenous factors, simply tell the candidate to assume that the market is expected to continue as it currently is.

*The candidate needs to calculate the value of the business now. This is a mathematical exercise.*

*Correct Answer:*

$$\begin{aligned} \text{Contribution Margin} &= \$200 / \text{coffin} \\ &\quad \times 40,000 \text{ coffins} \\ &\quad \times 10\% \text{ market share} \end{aligned}$$

$$= \$800,000$$

$$\begin{aligned}\text{Profit} &= \text{CM} - \text{Fixed Costs} \\ &= \$800,000 - \$700,000 \\ &= \$100,000\end{aligned}$$

Assuming a discount rate of 10% (*candidate can assume anything reasonable here as long as they are consistent later*) a perpetuity with cash flows of \$100k / year has a PV of  $\$100,000 / .1 = \$1\text{M}$ . So the current business is worth \$1M whether they keep it or sell it.

**Prompt: So now what is the value of the company if it were shut down and the assets were sold?**

*Information to give if asked:*

**Assets** – Since the firm has been building coffins by hand, the fixed assets are essentially only the land and improvements. These are owned outright by the company.

*When the candidate asks for the value of the land, have them brainstorm ways that they might determine this. They should come up with at least 3 good ways, such as:*

- Look for comparable real estate and determine recent selling price
- Find comparable commercial real estate and determine the rent per square foot, then discount the cash flows generated by renting the property
- Determine rate of appreciation for property in the area and then apply to book value of current land and improvements

*Give the candidate the following information and have them calculate the value of the property:*

**Book Value of Land:** \$20,000  
**Book Value of Improvements:** \$80,000  
**Years Owned:** 48  
**Avg. Real Estate Appreciation:** 6% / year

*Right Answer:* Using the “rule of 72,” a 6% growth rate will double the investment every  $72/6 = 12$  years. Since the property was held for 48 years, the current value will be  $\$100\text{k} * (2^4) = \$1.6\text{M}$ .

Since the assets (\$1.6M) are higher than the value of the discounted cash flows (\$1M), then it would make more sense to liquidate the business and sell the assets.

**Prompt: What would the value of the company be if he invests in the new technology?**

*Provide the following information if asked:*

**Investment** – Investing in the new technology will cost the firm \$1M.

**Cost Savings** – Material costs remain the same, but labor costs are reduced by 50%.

**Proprietary Nature of Technology** – The new coffin-making technology is being offered for sale by a machine tool company, who holds the patent. They are not offering exclusivity to any customers (i.e. they will sell to Maldovian Coffin's competitors if possible).

**Competitive Threat** – It is not known whether the competitors have acquired or are planning to acquire this technology.

**Customer Preferences** – While the machine-made coffins are not “hand made”, the quality perceived by the customer is the same or better. It is believed that the customer will be indifferent between the quality and appearance of a hand-made and a machine-made coffin.

**Brand Impact** – The candidate may argue that a machine-made coffin might negatively impact Maldovian Coffin's brand. If so, ask them how they would test this (e.g. consumer research), but tell them to assume that it would have negligible impact.

*Good Answer:* Since Maldovian Coffins has no proprietary control over the technology, it is likely that competitors will also acquire it, resulting in an overall lowering of the industry cost structure. If this is the case, price will also fall as competition cuts price in an attempt to gain share. If we assume that gross margins remain the same, since the industry competitive structure has not changed we can calculate the new margin contribution as follows:

$$\begin{aligned}\text{Gross Margin} &= \$200 / \$5,000 = 4\% \\ \text{Labor Cost} &= (4800 \times 90\%) \times 50\% = \$2,160 \\ \text{Material Cost} &= \$480 \\ \text{COGS} &= \$2,160 + \$480 = \$2,640\end{aligned}$$

$$\begin{aligned}\text{Price} &= \$2,640 / (1 - 4\%) = \$2,718 \\ \text{Contribution Margin} &= \$2,718 - \$2,640 = \$78 \\ \text{Loss} &= \$78 \times 4,000 - \$700,000 = -\$388,000\end{aligned}$$

So the introduction of the technology to the market might be expected to reduce industry profits, making this business completely unprofitable.

*Candidates could argue other scenarios, by assuming that the industry would be able to maintain higher margins than we have assumed here, so the answer may be different. They should recognize, however, that the introduction of this non-proprietary technology*

*will significantly reduce industry pricing in the absence of some other form of price support (such as branding, collusion between players, etc.)*

### **Conclusion**

*A star candidate will see that his/her time is nearly up and will present a recommendation for the client without prompting. If the interview is within 3 minutes of the end, ask:*

“The owner just called and said he has an offer to buy his business. He needs to know whether he should take it right now.”

*Good Answer:* Given the credible threat of the industry becoming unprofitable due to the introduction of this new technology, the owner should look to sell the company as soon as possible. Taking into account the assets of the firm and the present value of the expected cash flows of the business itself, he should attempt to liquidate the business and to sell the assets for around \$1.6M.

If unable to sell the business now, he can continue to operate the business as a cash cow, but should not invest in the business above what is necessary to keep it operating at its present level. He should expect the business to become less profitable as the industry moves to mechanization, and should eventually look to sell the assets of the company and close the firm.